# Do Now · retrieval

## A*: heuristic search I

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Why are predecessors recorded when a distance improves?

### Q2

Can Dijkstra assume negative edge weights are safe?

### Q3

What is 2+3+2?

### Q4

Starting S=0, edge S–A=4, then A–G=2. What is the S→A→G path cost?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

g, h and f have different meanings — g is distance already travelled; h estimates the remaining distance; f=g+h is the estimated total.

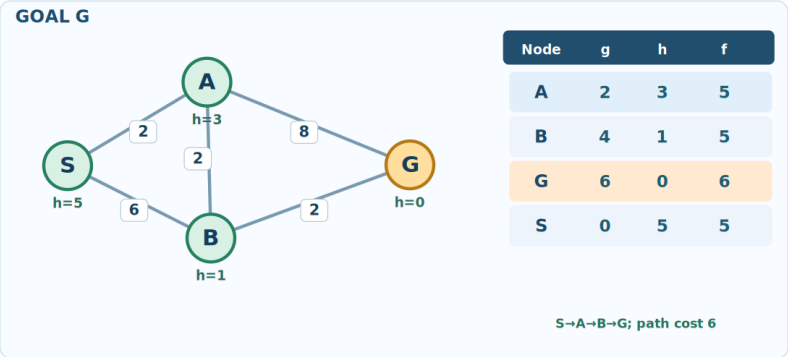
## Core idea 2

Start with a frontier — From S, A gets g2,h3,f5, while B gets g6,h1,f7. Choose A, not simply the smallest h.

## Core idea 3

Improve a discovered route — From A, B improves to g4,h1,f5. From B, G improves to g6,h0,f6.

**Key visual · complete concept**



# New Learning · applying the ideas

## Core idea 4

Stop on the selected goal — When G is selected from the frontier, reconstruct S→A→B→G with total cost 6.

## Core idea 5

Keep the heuristic safe — The estimates h(S)=5,h(A)=3,h(B)=1,h(G)=0 never exceed the actual cheapest remaining cost.

## Vocabulary · use the correct technical terms

g(n) — Known best cost from the start to vertex n.

h(n) — Estimated cost from vertex n to the goal.

f(n) — Combined score g(n)+h(n).

frontier / open set — Discovered vertices awaiting expansion.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

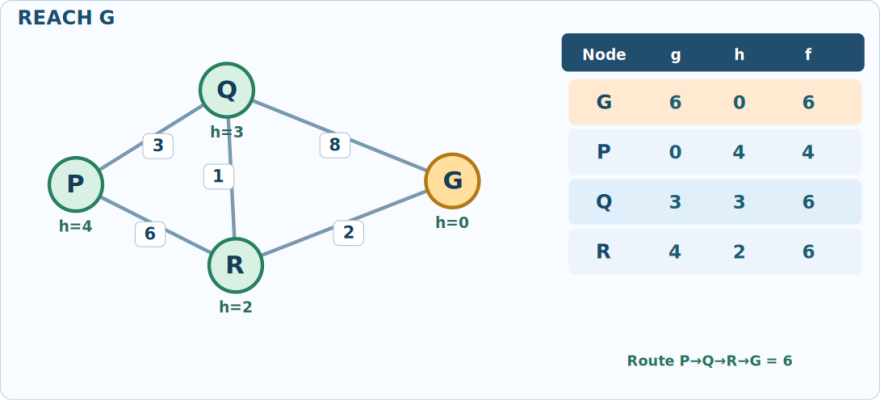
## Problem

Trace A* on P, Q, R, G using the supplied admissible estimates.

## Follow the worked method

1. Predict the next state — From P, Q has g3+h3=f6, R has g6+h2=f8. Choose Q.
2. Update the table — From R, G improves to g6,h0,f6.
3. Update the table — Select G: predecessor chain yields P→Q→R→G, cost 6.

**Source model · final illustrated step**




## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict the next state
From M, choose N because f(N)=2+3=5 < f(O)=5+1=6.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

A node has g=7 and h=3. Give f and define each value.

## Question 2

Graph edges S–A=2, S–B=5, A–B=1 and B–G=2. Heuristics: h(S)=4,h(A)=2,h(B)=1,h(G)=0. Calculate f for A and B when first found and state the optimal S→G route.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain why admissibility matters. For the graph S–A=2, A–G=4 and S–G=10, what is the greatest admissible h(A), and why?

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

A* chooses the next vertex solely by its smallest heuristic h(n).

## Use key vocabulary from memory

1. Explain g(n) in your own words.

2. Explain h(n) in your own words.

3. Explain f(n) in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.